Polydispersity Index of Polymers Revealed by DOSY

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The polydispersity index (PDI) of a polymer sample is a very important feature. It can reflect the purity of a standard sample or even modulate the physico-chemical properties of a polymer. The PDI is the ratio of the weight average molar mass (Mw) of the sample to its number average molar mass (Mn). NMR is a very useful tool to characterize polymers. Although ILT (Inverse Laplace Transform) analysis of DOSY (Diffusion-Ordered Spectroscopy) experiments was proposed to study mass distributions [1,2], no convincing method to measure PDI has been developed yet.

Mw can be evaluated from DOSY spectra via the fractal dimension [2,3]. On the other hand, it should also be noted that the terminal monomeric units of a polymer sample adopt the same statistical properties as Mn. When visible, their diffusion coefficient should be different from the other units. To test this hypothesis, alarge number of samples of Poly(Ethylene Oxide) (PEO) with different PDI was used for the study.

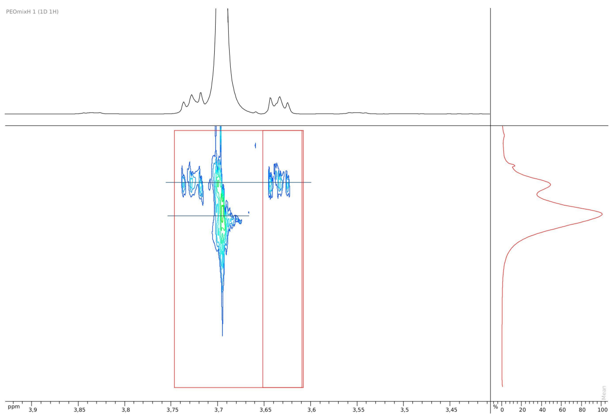

Figure 1. DOSY experiment on a PEO sample. Horizontal bars show the two different diffusion coefficients measured on the sample.

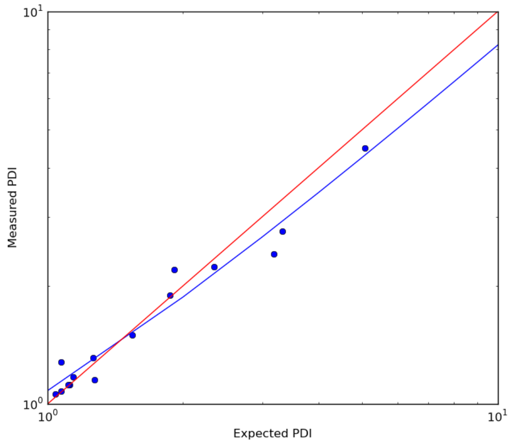

Figure 2. Correlation between measured and expected PDI (in blue) compared to the x=y equation (in red).

As expected, two diffusion coefficients can be measured on the same DOSY spectrum for a polymer sample. PDIs are then easily determined. The measured PDIs are quite close to the expected values. Even the standard sample low PDIs can be estimated precisely because the results of this method are completely independant of temperature or viscosity, they only depend on the fractal dimension.

Extending this method to block co-polymers and to polymers with no visible terminal monomeric units via chemical labelling will also be discussed.

[1] Chen et al. *J. Am. Chem. Soc.* 1995**, 117,** 7965-7970
[2] Håkansson et al. *Colloid Polym Sci* 2000, **278**, 399-405
[3] Augé et al. *J. Phys. Chem. B* 2009**, 113***,* 1914–1918